

Vacuum Cleaner AI Problem

Aim

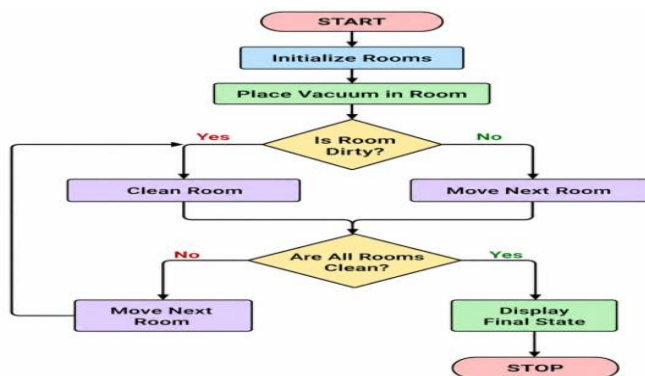
To design a simple AI-based vacuum cleaner agent that cleans all dirty rooms by sensing the environment and performing appropriate actions.

Objective

- To understand the concept of an intelligent agent.
- To simulate the working of a vacuum cleaner in AI.
- To clean all dirty rooms automatically.
- To minimize unnecessary actions.

Algorithm

1. Start.
2. Initialize the room states (Clean/Dirty).
3. Place the vacuum cleaner in the starting room.
4. Check the current room.
5. If the room is **Dirty**, clean it.
6. Move to the next room.
7. Repeat Steps 4–6 until all rooms are clean.
8. Display the final room states.
9. Stop.



PROGRAM

```

rooms = {
    "A": "Dirty",
    "B": "Dirty"
}

location = "A"

print("Initial State:")
print(rooms)

while True:

    if rooms[location] == "Dirty":
        print(f"\nVacuum is in Room {location}")
        print(f"Cleaning Room {location}")
        rooms[location] = "Clean"

    if location == "A":
        print("Moving to Room B")
        location = "B"
    else:
        print("Moving to Room A")
        location = "A"

    if rooms["A"] == "Clean" and rooms["B"] == "Clean":
        print("\nAll rooms are clean.")
        break

print("\nFinal State:")
print(rooms)

```

OUTPUT

```

>>> = RESTART: C:/Users/ATHIN/AppData/Local
ssssssssssss.py
Initial State:
{'A': 'Dirty', 'B': 'Dirty'}

Vacuum is in Room A
Cleaning Room A
Moving to Room B

Vacuum is in Room B
Cleaning Room B
Moving to Room A

All rooms are clean.

Final State:
{'A': 'Clean', 'B': 'Clean'}
>>> |

```

Result

The AI vacuum cleaner agent successfully sensed the environment, cleaned all dirty rooms, and stopped after completing the task. Thus, the Vacuum Cleaner AI Problem was implemented successfully using Python.